

Integrating External Knowledge with LLMs: A Systematic Review of RAG Approaches

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Abstract—Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation, exhibiting exceptional in-context learning abilities without requiring extensive fine-tuning. However, these models often suffer from significant limitations, including hallucinations where fabricated or incorrect information is presented as fact, and the need for constant retraining in order to effectively integrate new knowledge. Retrieval-Augmented Generation (RAG) has emerged as a promising paradigm to address these challenges by combining LLMs with information retrieval (IR) techniques. By leveraging external knowledge bases, RAG pipelines retrieve and incorporate relevant information dynamically, enhancing the factual accuracy and adaptability of LLMs. This paper provides an overview of the state-of-the-art methods for implementing RAG in LLMs, examining key components of the RAG pipeline, including data preparation, retrieval, reranking, and post-retrieval techniques. We aim to highlight how these components collectively enable LLMs to achieve greater reliability and flexibility, paving the way for more robust AI applications.

Keywords—RAG, Retrieval, IR, LLM

I. INTRODUCTION

The late 2010s saw a major shift in natural language processing (NLP) with the introduction of the transformer architecture [1]. Pretrained language models (LMs) [2], [3] significantly advanced the field, and subsequent scaling of these architectures to billions of parameters, trained on trillions of tokens, led to the emergence of large language models (LLMs) [4]–[6]. These models exhibit strong in-context learning, excelling in zero-shot and few-shot scenarios [7].

Despite their capabilities, LLMs face challenges such as hallucinations [8], where they generate factually incorrect or outdated information. Retrieval-Augmented Generation (RAG) [9] has emerged as a key approach to address these issues by integrating external knowledge into the model's context. By grounding responses in retrieved factual data, RAG enhances accuracy and mitigates hallucinations. Its effectiveness relies on a robust information retrieval (IR) system to extract relevant content from large knowledge corpora.

In this paper, we provide a comprehensive overview of the key components of the RAG pipeline. While previous work [10]–[12] provides an overview of the research area, this study gives more emphasis on the IR component in a concise and organized manner.

II. DATA PREPARATION

The first step in a RAG pipeline is preparing the knowledge collection for retrieval. Since LLMs have a limited context window, and research indicates that irrelevant content within this window degrades performance on downstream tasks [13], documents must be split into smaller, structured chunks—a process known as chunking.

A basic approach involves splitting text into fixed-size character chunks. A more effective method is recursive chunking, which respects document structure by iterating through priority separators (e.g. sections, paragraphs, sentences) until all chunks fit within the target size. Overlapping chunks help preserve context across splits.

Beyond structural chunking, semantic chunking uses embedding-based similarity to detect content shifts. Singh et al. [14] employ dense embeddings and dot product similarity to identify optimal chunk boundaries. However, Qu et al. [15] argue that semantic chunking is beneficial only when data exhibits sharp topical transitions.

Alternative strategies include proposition-based chunking, where Chen et al. [16] fine-tune a T5 model to extract atomic, self-contained factoids. LLM-assisted chunking, as proposed by Duarte et al. [17], prepends sentence IDs and prompts an LLM to identify context shifts, while DStar [18] enhances this by ranking chunks based on query similarity, preserving contextual relevance during retrieval.

For tables, converting them to JSON before chunking prevents structural loss. Images and graphs can be processed with image-to-text models to generate textual descriptions, ensuring multimodal information is preserved.

III. RETRIEVAL

Retrieval is the most critical component of the RAG pipeline. In this step, the user's query must be matched with documents or chunks containing relevant content. In the field of information retrieval (IR), this is known as Open-Domain Question Answering (ODQA) [19], and many research advancements in ODQA can be directly applied to RAG pipelines.

A. Retrieval Strategies

Various retrieval strategies can be employed to identify and retrieve documents relevant to a given query. This subsection provides a broad overview of the most widely used retrieval strategies in RAG.

1) *Sparse retrieval*: Sparse retrieval methods, such as term frequency–inverse document frequency (TF-IDF) and BM25 [20], are traditional techniques that represent documents as sparse vectors in a high-dimensional space, with each dimension corresponding to a term in the vocabulary. These methods are highly effective for exact term matching, making them suitable for scenarios where precise keyword overlap is crucial. However, they face limitations such as the vocabulary mismatch problem, where different words represent the same meaning, and the semantic mismatch problem, where the same word can have different meanings depending on the context.

Recent advancements leverage contextualized embeddings to enhance sparse retrieval. Dai et al. propose the DeepCT framework [21], which uses a BERT [2] model to compute term importance scores as weighting factors for traditional algorithms like BM25. Similarly, DeepImpact [22] calculates term "impact" scores to improve document ranking. The COIL framework [23], [24] incorporates contextual embeddings into the inverted index to capture semantic meaning and computes query-document similarity based on these embeddings.

Another prominent approach, SPLADE [25], [26], directly generates sparse representations using BERT. These representations are stored in the inverted index, allowing for more effective retrieval while maintaining sparsity.

2) *Dense retrieval*: Dense retrieval involves training language models to generate fixed-size vectors that represent both queries and passages. Reimers and Gurevych [27] demonstrated that models like BERT can be fine-tuned for this purpose using a bi-encoder architecture, which independently encodes queries and passages into n -dimensional vectors. During inference, all documents are pre-encoded and stored in an Approximate Nearest Neighbor (ANN) search index, such as FAISS [28], enabling efficient retrieval by embedding the query and finding its nearest neighbors. Training typically uses triplet or contrastive loss to optimize the distance between query embeddings and their corresponding passages.

Due to limited labeled data, unsupervised pretraining plays a critical role in reducing the need for extensive fine-tuning datasets. Lee et al. [29] introduced the Inverse Cloze Task (ICT), while subsequent works proposed tasks like Body First Selection (BFS) and Wiki Link Prediction (WLP) [30], Independent Cropping [31], and SimLM [32]. Additionally, Dai et al. [33] suggested leveraging large language models to generate synthetic datasets for training.

The selection of negative examples is another key aspect of training. Karpukhin et al. [34] proposed using in-batch negatives combined with BM25-derived relevance labels. Xiong et al. [35] introduced hard negatives derived from embeddings of previous model checkpoints stored in an ANN index. Hofstätter et al. [36] employed clustering with pretrained cross-encoder embeddings to obtain both hard and easy negatives, while Qu et al. [37] utilized cross-batch negatives. Izacard et al. [31] leveraged stored embeddings from separate model instances as negative examples.

Despite their efficiency, dense retrieval models are constrained by the need to represent all relevant information in a single vector, which limits their performance compared to cross-encoders (see III-B) and multi-vector dense models (see III-A3). To address this, several works [32], [37], [38] propose using these more capable models as teachers in knowledge distillation tasks.

Dense retrieval remains an active research area with numerous models available. Open-source implementations, such as [39], and proprietary ones, like [40], offer a range of options. The MTEB [41] leaderboard¹ is a valuable resource for identifying suitable models for specific tasks.

3) *Multi-vector dense retrieval*: Multi-vector dense retrieval (MVDR) methods enhance the expressiveness of retrieval systems by representing queries and documents as multiple token-level vectors instead of single fixed-length embeddings. This addresses the limited capacity of fixed-length representations.

ColBERT [42] employs a BERT-based model to generate token embeddings and identifies relevant documents through a per-token MaxSim operation. However, this approach incurs substantial memory overhead due to the storage of token embeddings. Santhanam et al. [43] optimize memory usage by clustering embeddings and storing cluster IDs with residuals instead of full token embeddings, while also distilling knowledge from a cross-encoder teacher model to enhance performance. PLAID [44] further improves efficiency by leveraging centroids to identify candidate documents, only expanding them to their full embeddings during the final ranking stage.

Zhou and Devlin [45] introduce the Multi-Vector Attention Network (MVA), which replaces the MaxSim operation with an attention mechanism to aggregate scores. Qian et al. [46] propose the ALIGNER framework, which utilizes a sparse alignment matrix and unary salience scores to enable multiple document tokens to attend to a single query token, significantly reducing the number of stored tokens.

While MVDR methods deliver superior retrieval performance compared to single-vector approaches, they come with increased computational and memory requirements, particularly in terms of storage overhead.

4) *Hybrid retrieval*: Hybrid retrieval methods aim to address the limitations of sparse and dense retrieval. Sparse retrieval methods, such as BM25, suffer from vocabulary and semantic mismatches, while dense retrieval struggles with exact term-based matching. Research indicates that combining these approaches improves retrieval effectiveness [34], [47], [48].

Score fusion is commonly performed using either a convex combination of scores or Reciprocal Rank Fusion (RRF). RRF has the advantage of being independent of scoring scales, as it relies only on rank positions, while convex combination has been found to be more robust and often yields better performance [49].

¹<https://huggingface.co/spaces/mteb/leaderboard>

Recent efforts aim to unify sparse and dense retrieval within a single framework. Gao et al. propose CLEAR [48], where a dense retriever is trained to complement BM25, and their scores are combined using a convex combination at inference time. Similarly, Chen et al. introduce SPAR [50], in which a dense retriever is trained to mimic BM25 labels. The resulting vector representations from both standard dense and synthetic BM25 retrievers are then merged, enabling retrieval solely via an approximate nearest neighbor (ANN) index.

5) *Generative retrieval*: Traditional retrieval follows an index-then-retrieve paradigm, where indexing and retrieval are separate processes. Generative Retrieval (GR) introduces a unified, end-to-end approach, enabling faster retrieval while addressing the expressivity limitations of fixed-vector encodings. By generating document identifiers directly from model parameters, GR allows for richer and more nuanced representations.

The Differentiable Search Index (DSI) framework introduced in [51] employs a transformer-based sequence-to-sequence model that takes a query as input and generates a document ID (docid) as output. The docids are represented using various formats: atomic identifiers, naive string identifiers, and semantically structured identifiers. The model is trained to map document content (and sometimes queries) to the corresponding docids.

Subsequent advancements have refined this framework. For instance, DSI-QG [52] improved DSI by training models solely on queries generated via a fine-tuned docT5query model [53]. Wang et al. [54] enhanced semantically structured identifiers by differentiating output tokens based on their positional importance within the docid. Their PAWA method uses an auxiliary transformer to generate token-specific transformation matrices for docid generation. Bevilacqua et al. proposed Search Engines with Autoregressive LMs (SEAL) [55], which leverages n-grams as docids and employs FM-index [56] for efficient constrained decoding. Similarly, Li et al. introduced MINDER [57], which extends on SEAL and utilizes multiple docid representations (titles, document slices, and pseudo-queries) and aggregates scores across these identifiers at inference. Their later work, LTRGR [58], addresses the mismatch between training (sequence generation) and inference (document ranking) by incorporating a fine-tuning step with a margin-based ranking loss. RIPOR [59] further optimizes the token-by-token generation process by ensuring relevance is reflected throughout the sequence generation, aligning loss calculation with inference mechanisms like beam search.

Despite its promise, GR currently lags behind dense retrieval methods in terms of performance. Key challenges include scaling to larger datasets [60] and efficiently updating the index with new documents without catastrophic forgetting [61]. Recent advancements, such as the unification of GR and MVDR into a single framework [62], offer promising directions of research.

B. Reranking

Reranking refines and filters a set of documents retrieved in the first-stage retrieval process. Since this document set is significantly smaller than the full corpus, computationally expensive cross-encoding methods can be employed, where both the query and document tokens are simultaneously processed to improve relevance predictions.

Nogueira et al. [63] propose concatenating the query with each document and using a BERT model's [CLS] token to compute relevance scores. They further introduce duoBERT [64], which applies an additional reranking step by comparing document pairs with respect to the query in a tournament-style evaluation. Zhuang et al. [65] adapt the T5 [3] transformer for reranking using two approaches: a full encoder-decoder model with a special ranking token and an encoder-only method pooling token representations for scoring, similar to [63]. Sachan et al. [66] explore prompting pre-trained language models (PLMs) to estimate query likelihood given a document, using this probability as a relevance signal.

The advent of large language models (LLMs) has enabled the use of extended context windows and strong zero-shot performance. Pradeep et al. [67] fine-tune an LLM for listwise reranking, where the model directly outputs an ordered list of document indices. Zhuang et al. [68] demonstrate that prompting LLMs to assign numerical relevance scores outperforms binary relevance judgments. Ma et al. [69] show that simple listwise ranking prompts yield strong performance without fine-tuning.

Reranking is a crucial step in retrieval pipelines, significantly improving relevance compared to first-stage retrieval. However, the computational cost of reranking must be carefully balanced against latency constraints, especially in real-time applications.

C. When and what to retrieve?

In the standard RAG approach, after each user's query, relevant documents are retrieved, and an answer is generated. However, some research explores more adaptive methods to determine when and what to retrieve.

Jiang et al. propose Forward Looking Active RAG (FLARE) [70], which is based on the assumption that LLMs are well-calibrated and that low-probability token generation indicates a lack of knowledge. When such a token is generated, the next sentence is used for retrieval, and the generation process is backtracked to insert the retrieved content at the low-confidence token's position. Yan et al. introduce Corrective RAG (CRAG) [71], where a T5 transformer evaluates retrieval quality, similar to a reranker. If relevant documents are found, the retrieval results are refined and presented to the model; otherwise, a web search is performed instead. Shao et al. [72] propose iterative retrieval, where each subsequent retrieval incorporates context from previous retrievals, functioning similarly to chain-of-thought prompting [73]. Wang et al. [74] explore a classifier-based approach to determine

whether the model can answer a query using its parametric knowledge or if retrieval is necessary. Asai et al. propose Self-RAG [75], where special tokens are embedded in the LM's output to trigger retrieval dynamically and reflect on the retrieved results as needed.

D. Query and Document Expansion

Query and document expansion enhance retrieval performance by augmenting either the query or the document with additional information. Query expansion involves reformulating or extending the query, while document expansion enriches documents with supplementary content.

Nogueira et al. [53] fine-tune a T5 model to generate questions that a document answers and append them to the document for improved retrieval. Mao et al. [76] employ a language model (LM) for unsupervised query expansion using diverse contexts. Gao et al. [77] propose the Hypothetical Document Embeddings (HYDE) method, where a large language model (LLM) generates N hypothetical documents that answer the query. These documents are embedded using a dense retriever, and their mean vector represents the query. Similarly, Wang et al. [78] generate a hypothetical document using an LLM but instead append it to the original query for retrieval. To mitigate the computational cost of LLM-based expansion, some approaches [79], [80] leverage reinforcement learning and relevance signals to fine-tune query rewriting models, achieving competitive or superior performance.

While these methods effectively improve retrieval [81], their added latency must be considered. Additionally, studies indicate that query rewriting can degrade performance when used with cross-encoder document rerankers [82], [83].

E. Data organization

Research has also explored structured representations for storing corpus knowledge. Trajanoska et al. [84] propose leveraging LLMs to generate knowledge graphs. Edge et al. [85] extend this approach by applying community detection algorithms to these graphs, creating content summaries to enhance global data understanding.

Liu et al. [86] introduce a hierarchical tree structure that decomposes individual documents and employs a two-stage retrieval process: first, using document summaries to identify relevant documents, and second, retrieving specific content within them. Similarly, Sarthi et al. [87] construct a tree-like structure at the corpus level, clustering similar nodes under parent nodes and summarizing their content. This approach facilitates accessing both high-level knowledge aggregated in the parent nodes and fine-grained retrieval through lower-level child nodes.

IV. POST RETRIEVAL

After retrieval, it is crucial to present the retrieved documents in a format optimized for the LLM. In addition to standard prompting techniques, such as using markdown

for content separation, studies have highlighted the "lost in the middle" problem [88], where information placed in the middle of the context window is less effectively utilized than content at the beginning or end. To mitigate this issue, it has been proposed to reorder documents so that the most important ones appear at the start and end, while less relevant documents are placed in between [89].

V. CONCLUSION

Retrieval-augmented generation (RAG) is a powerful paradigm that leverages LLMs' language capabilities to summarize content and answer user queries. A high-quality RAG pipeline relies on accurate retrieval pipeline. This paper provides a broad overview of key techniques used in RAG, many of which are implemented in libraries such as LangChain², LangGraph³, and LlamaIndex⁴, facilitating streamlined pipeline development. Effective prompting strategies can further improve RAG performance [90].

Due to space constraints, we do not cover real-life applications or end-to-end training of RAG systems and refer readers to [11], [12] for detailed coverage.

Given the rapid advancement of AI and LLMs, some techniques described in this paper may quickly become outdated. Continuous experimentation with these methods is essential to assess their relevance and effectiveness in specific RAG applications.

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²<https://www.langchain.com/>

³<https://www.langchain.com/langgraph>

⁴<https://www.llamaindex.ai/>

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